



AEO “Nazarbayev Intellectual Schools”

CHEMISTRY

Grade 12

Paper 2

May 2025

MARK SCHEME

Maximum Mark: 100

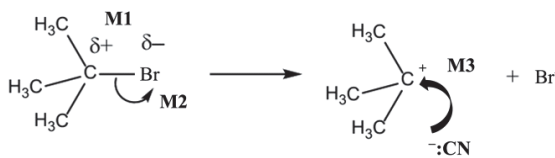
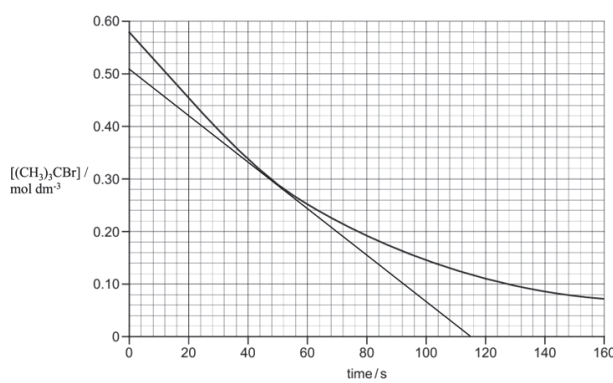
This document consists of **10** printed and **2** blank pages.

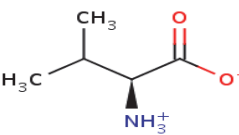
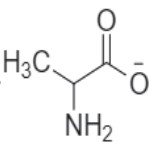
ANNOTATIONS AND ABBREVIATIONS

line space	new marking point for same question item
/	meaning “or” indicates alternative correct responses for same marking point
() brackets	the words or units in brackets do not need to be stated but must be implied For example, 10 (J) means that the mark is awarded for “10” but not for “10 kJ”
<u>underlining</u>	exact word is required
accept	Indicates a less than perfect, but in this case, acceptable response.
allow	identifies an answer that will be allowed if no other marking point is awarded.
not	Indicates an incorrect response that would contradict another otherwise correct alternative i.e. right plus wrong penalty applies.
ignore	Indicates an incorrect response which is irrelevant and does not cause a right plus wrong penalty.
ecf	meaning "error carried forward", where a wrong response in an earlier part of a question is used correctly in a later part of the same question
sig. figs	Answers are normally acceptable to any number of significant figures ≥ 2 . This indicates an exception where a specified number of significant figures is required. Numbers must be correctly rounded and not truncated.
note	often added for in additional guidance to help the exam marker.
con	this is used when marking exam papers to indicate that a learner has given a contradictory answer and loses a mark.
ora	or reverse argument
owtte	or words to that effect
AVP	any valid point

PRE-STANDARDISATION

Question	Answer	Mark	Additional guidance
1 (a)	(Atomic radius) shows a general <u>decrease</u> ;	1	Allow same number of shells
	The number of protons increases OR nuclear charge increases;	1	
	Shielding is similar/same OR electrons are added to the same shell.	1	
		[3]	
1 (b) (i)	Al^{3+} , Mg^{2+} , Na^{+}	[1]	
1 (b) (ii)	Mg^{2+} contain fewer electrons than protons	1	OWTTE
	Nucleus has a greater attraction	1	
		[2]	
1 (c) (i)	Enthalpy/energy (needed) to remove one mole of electrons from one mole of gaseous atoms;	1	
	To form one mole of gaseous 1+ ions / ions with +1 charge	1	
		[2]	
1 (c) (ii)	$\text{P(g)} \rightarrow \text{P}^{+}(\text{g}) + \text{e}^{-}$ State symbols must be included for the mark	[1]	Allow $\text{P(g)} - \text{e}^{-} \rightarrow \text{P}^{+}(\text{g})$
1 (c) (iii)	Cross above 1012	[1]	
1 (c) (iv)	Electrons are now being removed from positive ions	[1]	
2(a)	✓ ☐ rate of forward reaction = rate reverse reaction	1	
	✓ ☐ macroscopic properties remain constant/ concentrations remain constant	1 [2]	
2(b)(i)	$K_w = [\text{H}^{+}][\text{OH}^{-}]$ $\text{mol}^2 \text{ dm}^{-6}$	1 1 [2]	
2(b)(ii)	K_w will be higher in high temperature	1	
	because reaction is endothermic	1 [2]	
2(c)(i)	$[\text{H}^{+}] = (1 \times 10^{-14}) / 1 \times 10^{-2} = 1 \times 10^{-12}$	1	ecf correct ans = [2]
	$\text{pH} = -\log_{10}[\text{H}^{+}] = 12$	1 [2]	

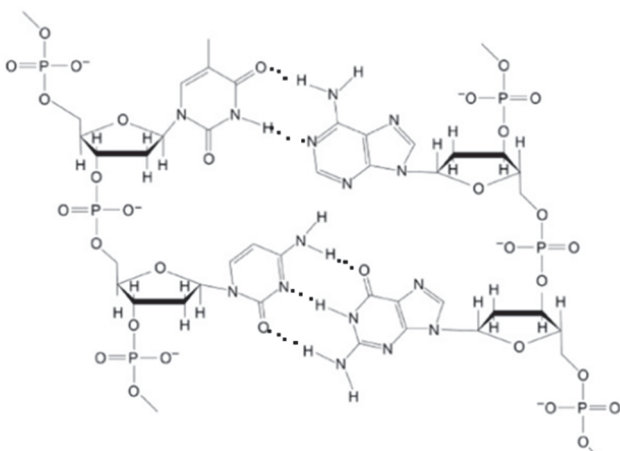
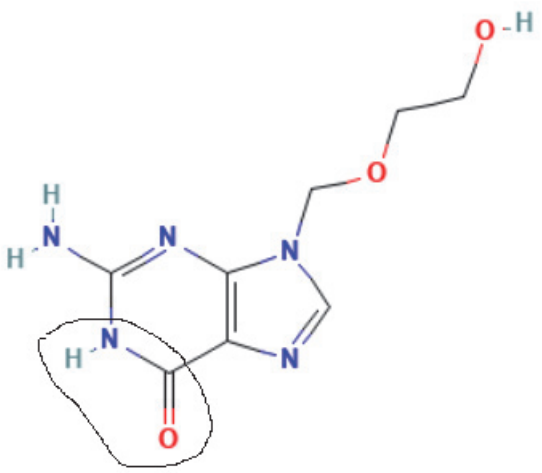
2(c)(ii)	$[\text{CH}_3\text{NH}_3^+][\text{OH}^-] / [\text{CH}_3\text{NH}_2]$	[1]	Do not allow $[\text{CH}_3\text{NH}_3^+][\text{OH}^-] / [\text{CH}_3\text{NH}_2][\text{H}_2\text{O}]$
2(c)(iii)	$[\text{OH}^-] = \sqrt{4.38 \times 10^{-4} \times 0.010}$ $[\text{OH}^-] = 2.09 \times 10^{-3}$ $[\text{H}^+] = 4.78 \times 10^{-12}$ pH = 11.3	1 1 1 1 [4]	Allow M1 for correct calculation of $[\text{OH}^-]$
2(c)(iv)	Methylamine is partially dissociated/ionised OR KOH is completely dissociated/ionised	[1]	OWTTE
3(a)	reagent that forms bonds by donating an electron pair	[1]	Accept lone pair
3(b)	 M1: dipole on C–Br bond AND curly arrow breaking C–Br bond M2: tertiary carbocation and Br[−] (ion) formed in the mechanism M3: arrow from the lone pair of electrons on the carbon atom of the negatively charged cyanide ion to the central C⁺ atom.	1 1 1 [3]	Max 1 mark for the wrong reactant or “sticks”
3(c)(i)		[1]	Accept any appropriate tangent. Line must touch the curve at 0.30 but must not cross the curve.
3(c)(ii)	Stage 1: Rate at concentration = 0.30 mol dm^{-3} from the tangent is $0.52/115-0=4.52 \times 10^{-3}$ Stage 2: Comparison of rates and concentrations Initial rate / rate at 0.30 = $(8.81 \times 10^{-3}) / (4.52 \times 10^{-3})$	1	Range between 4.50×10^{-3} and 4.80×10^{-3}

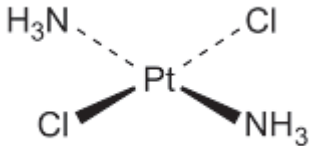
	$^3) = 1.95$ Initial concentration / concentration at point where tangent drawn = $0.58 / 0.30 = 1.93$ Stage 3: Deduction of order If order is 1, rate should increase by factor of $(1.93)^1 = 1.93$ this is approximately equal to 1.95 therefore order is 1 st with respect to 2-bromo-2-methylpropane	1 1 1 [4]	Marking points in stage 2 can be in either order
3(c)(iii)	Rate = $[(CH_3)_3CBr]$ k	[1]	
3(c)(iv)	Rate calculated from tangent/ 0.30 AND s^{-1}	[1]	ecf
3 (d)	Reaction with 2-iodo-2-methylpropane C-I bond energy is lower than C-Br / C-I bond is weaker than C-Br	1 1 [2]	
4(a)(i)	Condensation	[1]	
4(a)(ii)	$NH_2-CH(CH_3)-CO-NH-CH(C_3H_7)-COOH$	[1]	Accept: val-ala not polymers
4(a)(iii)		[1]	allow $-CO_2^-$
4(a)(iv)		[1]	
4(b)	☒ <u>ionic</u> attractions/bonds between $-CO_2^-$ and $-NH_3^+$ ☒ <u>van der Waals</u> between <u>alkyl / non-polar groups</u> ☒ <u>hydrogen(H) bonding</u> between $-OH$, $-NH_2$, $COOH$, $-NH$ ☒ <u>disulfide bridges</u> (cysteine residues/ S-S bonds)	1 1 1 [3]	Four correct answers -3 marks Three correct answers-2 marks Two correct answers -1 mark
4(c)(i)	Molecules with the <u>same structure/structural formula</u> :	1	

	But with bonds/atoms/groups <u>arranged differently in space</u> ;	1 [2]	
4(c)(ii)	Plane polarised light; Will be rotated in opposite directions;	1 1 [2]	
5(a)	Cl ₂ AND (UV) light/heat	[1]	Must be both answers to get mark
5(b)	Reagent Cl ₂ AND Catalyst AlCl ₃ /FeCl ₃ /Fe, etc.	[1]	Must be both answers to get mark
5(c)	M1 acidified potassium dichromate or K₂Cr₂O₇ / H₂SO₄ OR K₂Cr₂O₇ / H⁺ OR acidified K₂Cr₂O₇ M2 (orange to) green solution OR goes green M3 (solution) remains orange or no reaction or no (observed) change Alternative using KMnO₄ / H₂SO₄ M1 acidified potassium manganate(VII) / potassium permanganate or KMnO₄ / H₂SO₄ OR KMnO₄/ H⁺ OR acidified KMnO₄ M2 colourless solution OR goes colourless M3 (solution) remains purple or no reaction or no (observed) change	1 1 1 [3]	If no reagent or incorrect reagent in M1, no marks for M1, M2 and M3 If incomplete / inaccurate attempt at reagent e.g. “dichromate” or “dichromate(IV)” or incorrect formula or no acid, penalise M1 only and mark on For M2 ignore dichromate described as “yellow” or “red” For M3 ignore “nothing (happens)” or “no observation” For M1 If incomplete / inaccurate attempt at reagent e.g. “manganate” or “manganate(IV)” or incorrect formula or no acid, penalise M1 only and mark on Credit alkaline KMnO₄ for possible full marks but M2 gives <u>brown precipitate</u>
5(d)	Nucleophilic substitution	[1]	both words required. Accept phonetic

			spelling
5(e)	A mixture of liquids is heated AND Vapour is formed condensed back	1	
	To ensure that any alcohol and aldehyde is completely oxidised.	1 [2]	
5(f)(i)	(nucleophilic) addition-elimination	[1]	Not electrophilic addition-elimination Ignore esterification
5(f)(ii)	M1 arrow from the oxygen O-H of the alcohol group to the carbon atom of the carbonyl group and arrow from the double bond C=O group to the oxygen of the carbonyl group M2 Formation of a tetrahedral intermediate in which carbon is bonded to the oxygen of ethanol, chlorine and oxygen of the carbonyl group M3 final formula of the ester M4 3 arrows in the intermediate	1 1 1 1 [4]	extra curly arrows are penalized M2 not allowed independent of M1, + rather than δ^+ on C=O loses M2 If Cl lost with C=O breaking, max1 for M1
6(a)(i)	$2\text{Zn}(\text{NO}_3)_2(\text{s}) \rightarrow 2\text{ZnO}(\text{s}) + 4\text{NO}_2(\text{g}) + \text{O}_2(\text{g})$ correct formulae AND state symbols correct balancing	1 1 [2]	
6(a)(ii)	$2.16/81.4 = 0.0265$ Penalise if not 3 sig figs in this clip only	[1]	
6(a)(iii)	$0.0265 \times 2.5 = 0.0663$	[1]	Ecf
6(b)	M1: $pV = nRT$ $V = \frac{0.506 \times 8.31 \times 357}{100\,000} = 0.0150 \text{ m}^3$ 15.0 (dm ³) All 2 marks for 15.0 (dm ³)	1 1 [2]	However if 15.0 m³ or cm³ allow 1 (ie penalise wrong units in final answer)

6(c)(i)	$\text{ZnO} + 2\text{HNO}_3 = \text{Zn}(\text{NO}_3)_2 + \text{H}_2\text{O}$	[1]	Must be balanced
6(c)(ii)	$0.0247 \times 2 = 0.0494$	[1]	Allow 0.0500
6(c)(iii)	1.08 mol dm^{-3}	[1]	Ecf
7(a)	Strong absorption at $3580\text{-}3650\text{cm}^{-1}$	[1]	Accept peak at $3580\text{-}3650\text{cm}^{-1}$
7 (b) (i)	5	[1]	
7 (b) (ii)	Peak at $\delta = 3.63$ is $-\text{CH}_2$ next to OH AND Peak at $\delta = 0.97$ is CH_3 next to CH_2	[1]	
7 (b) (iii)	$-\text{CH}_2$ group is connected closer to more electronegative element/oxygen	[1]	
7 (b) (iv)	$\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-CH}_2\text{-OH}$	[1]	
7 (c) (i)	$\text{C}_4\text{H}_9\text{OH} + 6\text{O}_2 = 4\text{CO}_2 + 5\text{H}_2\text{O}$	[1]	
7 (c) (ii)	$q = 50.0 \times 4.18 \times 15.3 = 3198 \text{ (J) OR } 3.198 \text{ (kJ)}$ OR $3.20 \text{ (kJ) OR } 3.2 \text{ (kJ)}$ $n = 0.190 / 74.0 = 0.00257 \text{ mole}$ $\Delta H = -1244 \text{ kJ mol}^{-1}$	1 1 1 [3]	Penalise M3 ONLY if correct numerical answer but sign is incorrect. Therefore +1244 gains two marks. ecf for M3 from M1 and/or M2
7(c)(iii)	any two from <u>heat</u> loss or not all <u>heat</u> transferred to the apparatus or <u>heat</u> absorbed by the apparatus or (specific) heat capacity of the apparatus not considered - incomplete combustion / not completely burned / reaction is not complete - reactants and / or products may not be in standard states	1 1 [2]	
7(d)(i)	$\Delta H = \Sigma \Delta H \text{ breaking bonds} - \Sigma \Delta H \text{ forming bonds}$ OR correct cycle $\Delta H = [8536] - [10520]$ $\Delta H = -1984 \text{ (kJ mol}^{-1}\text{)}$ This answer only.	1 1 1 [3]	Correct answer with no working scores 1 mark only. M2 also implies M1 scored. Allow 1 mark out of 3 for correct value with incorrect sign.
7(d)(ii)	Bond energies calculated by using enthalpy changes of gaseous compound. Enthalpy change for combustion reaction is	1 1	OWTTE

	obtained by using liquid compound / the bond energy for C=O bond in the data booklet is an average value / bond energy value for C=O in data booklet is not the value for the C=O bond in carbon dioxide.	[2]	
8(a)(i)	 2 hydrogen bonds shown between adenine and thymine 3 hydrogen bonds shown between guanine and cytosine	1 1 [2]	
8(a)(ii)	Thymine and adenine AND cytosine and guanine.	[1]	
8(b)(i)		[1]	
8(b)(ii)	the acidity of the stomach before it can be absorbed into the blood/ Damage to the drug molecules at very low pH	[1]	
8(b)(iii)	Headache/ nausea/ vomiting/ abdominal pain and Stop taking the drug and see a doctor	[1]	

8 (c) (i)	Increase in number of molecules/ 2 moles to 3 moles so entropy change/ ΔS is positive similar type of bonds broken and formed so enthalpy change ΔH is approximately zero / negligible Free-energy change ΔG is negative	1 1 1 [3]	OWTTE
8(c)(ii)	Tumours have thin blood vessels with holes large enough So nanoparticle can pass into a tumour	1 1 [2]	ORA
8 (d) (i)		[1]	
8(d) (ii)	Cl ⁻ ligands located in opposite direction so can not form crosslink with DNA	[1]	

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